

Causes of Hallucination in Large Language Models

Introduction

Hallucination in large language models refers to the generation of text that is fluent and confident but unsupported, inaccurate, or inconsistent with available evidence. A hallucinated answer may contain fabricated facts, incorrect dates, wrong entity relationships, invented citations, or misleading explanations.

Large language models can produce hallucinations because they generate text by predicting likely continuations rather than directly checking truth against a verified knowledge source. Even when a model has seen relevant information during training, it may recall facts incorrectly or combine facts from different contexts.

Hallucination detection and correction are important for question answering, education, healthcare, law, finance, software engineering, and research assistance. In these settings, an answer that sounds plausible but is wrong can be more harmful than an answer that simply says the information is unavailable.

Background

Language models learn statistical patterns from large text collections. During generation, the model selects tokens based on learned probability distributions and the prompt context. This process can produce coherent text without guaranteeing factual accuracy.

Hallucination became a widely discussed issue as large generative models became common in public applications. Models such as GPT-style systems, instruction-tuned chatbots, and open-source local models can all hallucinate when evidence is missing, ambiguous, outdated, or incorrectly interpreted.

Retrieval-augmented generation was introduced as one practical method to reduce hallucination. Instead of relying only on internal model memory, a RAG system retrieves relevant external documents and uses them as grounding context for generation.

Core Types of Hallucination

Factual hallucination occurs when a model states an incorrect fact. For example, a model may say that a method was introduced in 2021 when the available evidence says it was introduced in 2020.

Entity hallucination occurs when a model invents or substitutes names, organizations, places, or people. For example, it may attribute a paper to the wrong authors or claim that one person won an event when another person did.

Numerical hallucination occurs when a model generates incorrect numbers, dates, percentages, or quantities. This includes wrong years, wrong benchmark scores, wrong dataset sizes, or fabricated statistics.

Citation hallucination occurs when a model invents references, paper titles, URLs, or source identifiers. This is common when the prompt asks for academic or legal references and the model tries to produce plausible-looking citations.

Instruction hallucination occurs when the model adds unsupported steps, assumptions, or explanations that were not requested or not supported by evidence. This can happen in technical troubleshooting or procedural answers.

Major Causes

One cause of hallucination is incomplete or missing evidence. If the prompt asks about a specific fact and the model does not have reliable context, it may still generate a confident answer instead of refusing.

Another cause is outdated training data. A model trained before a recent event may not know current facts. If it attempts to answer anyway, it may provide outdated or guessed information.

A third cause is ambiguity in the user query. If a term has multiple meanings, the model may choose the wrong interpretation. For example, an acronym can refer to different concepts in different domains.

Noisy or irrelevant retrieved context can also cause hallucination. In a RAG system, retrieval errors can pass unrelated evidence to the generator. The generator may then combine unrelated facts into a misleading answer.

Overgeneralization is another cause. The model may apply a pattern that is often true in training data but not true for the specific question. This is common when the answer involves dates, named entities, or exceptions.

Hallucination in RAG Systems

Retrieval-augmented generation reduces hallucination by grounding answers in retrieved documents, but it does not remove the problem completely. A RAG system can still hallucinate if retrieval fails, if the retrieved evidence is weak, or if the generator ignores evidence.

One common RAG failure is unsupported answer generation. The retriever may return passages related to the broad topic but not the specific fact being asked. For example, evidence may explain what RAG is but not state when it was introduced.

Another failure is evidence misinterpretation. A retrieved passage may contain the correct fact, but the model may summarize it incorrectly or combine it with prior knowledge in an unsupported way.

Citation misuse is also possible. A generated answer may include citation markers after claims that are not actually supported by the cited evidence. Therefore, citation grounding must be checked at the claim level rather than assumed to be correct.

Detection Methods

Claim extraction is a common first step in hallucination detection. The answer is split into factual claims, and each claim is checked against retrieved evidence. This is more precise than evaluating the entire answer as one block.

Semantic similarity can estimate whether a claim is related to evidence, but similarity alone does not guarantee factual correctness. A claim saying 2021 and evidence saying 2020 may still have high semantic similarity because most words overlap.

Factual consistency checks compare specific values such as years, dates, numbers, and entities. These checks can detect direct contradictions that similarity models may miss.

Natural language inference can classify whether evidence entails, contradicts, or is neutral toward a claim. NLI is useful for factual verification, but it must be used carefully because some NLI models classify unrelated statements as contradiction.

Span-level highlighting identifies the exact part of a claim that appears unsupported or contradicted. For example, in the sentence RAG was introduced in 2021, the span 2021 can be highlighted if the evidence does not support that year.

Correction Methods

Hallucination correction rewrites an answer using only supported evidence. The goal is not simply to shorten the answer, but to preserve supported claims, remove unsupported claims, and replace incorrect details with evidence-backed information.

A safe correction system should refuse to answer when evidence is insufficient. If the knowledge base does not support a specific claim, the system should say that the evidence is unavailable rather than inventing a fact.

Correction should be followed by re-verification. A corrected answer can still contain unsupported claims if the generator introduces new information during rewriting. Post-correction verification helps ensure that the final answer is safer than the original answer.

Evidence citations can improve transparency, but citations must be meaningful. A citation marker should appear only after a claim that is actually supported by the referenced evidence.

Evaluation

Hallucination evaluation can measure support ratio, hallucination rate, factual improvement, and correction success. Support ratio measures the proportion of claims that are supported by evidence. Hallucination rate measures the proportion of claims that are unsupported or contradicted.

Precision, recall, and F1 score can be used when a labeled dataset is available. Precision measures how many detected hallucinations are truly hallucinations. Recall measures how many actual hallucinations were detected. F1 combines precision and recall.

Faithfulness measures whether the generated answer stays grounded in retrieved evidence. Completeness measures whether the answer covers the important evidence without omitting key supported information.

Span-level evaluation can compare predicted hallucinated spans with gold spans. This is useful when the system highlights exact unsupported words, numbers, or entities.

Real-world Applications

In education, hallucination detection can help prevent students from receiving confident but incorrect explanations. A system can show which claims are supported and which require verification.

In healthcare and law, hallucination detection is critical because unsupported claims can have serious consequences. A safe system should clearly separate evidence-backed information from uncertain information.

In software engineering, hallucination detection can identify invented APIs, wrong configuration steps, and unsupported debugging advice. Evidence-grounded correction can make technical assistants more reliable.

In academic research, hallucination detection can reduce fabricated citations, incorrect summaries, and misleading claims about papers or benchmarks.

Limitations

Hallucination detection is difficult because factual correctness depends on evidence coverage. If the knowledge base lacks relevant evidence, the system cannot prove or disprove the claim. In such cases, safe refusal is better than unsupported generation.

Natural language inference models are not perfect. They can misclassify unrelated claims as contradiction or fail to recognize subtle entailment. A robust system should combine NLI with retrieval relevance, rule-based checks, and thresholding.

PDF extraction noise can also harm hallucination detection. Broken sentences, tables, page headers, references, and equation fragments can create poor chunks and irrelevant retrieval results.

Correction systems may accidentally introduce new hallucinations. This is why a corrected answer should be checked again before being shown as final output.

Future Directions

Future systems may use adaptive retrieval depth, where more evidence is retrieved when confidence is low. This can improve answerability decisions for difficult queries.

Better confidence calibration can help users understand whether a claim is fully supported, partially supported, unsupported, or contradicted. Calibrated scores are important for trustworthy deployment.

Stronger NLI models and domain-specific fact verifiers can improve contradiction detection. However, they must remain efficient enough for practical use on local hardware.

Web-scale retrieval and real-time knowledge updating can reduce outdated answers. Multimodal hallucination detection may extend these ideas to images, tables, charts, and audio.

Conclusion

Hallucination in large language models is a major reliability challenge because fluent answers can contain unsupported or false information. The problem is especially serious when users rely on models for factual, technical, or high-stakes tasks.

RAG reduces hallucination by retrieving external evidence, but retrieval alone is not enough. A strong system should verify claims, check factual consistency, highlight unsupported spans, correct unsafe answers, and refuse when evidence is insufficient.

Combining hybrid retrieval, claim-level verification, optional NLI, span highlighting, and post-correction checking provides a practical path toward safer and more trustworthy language model applications.